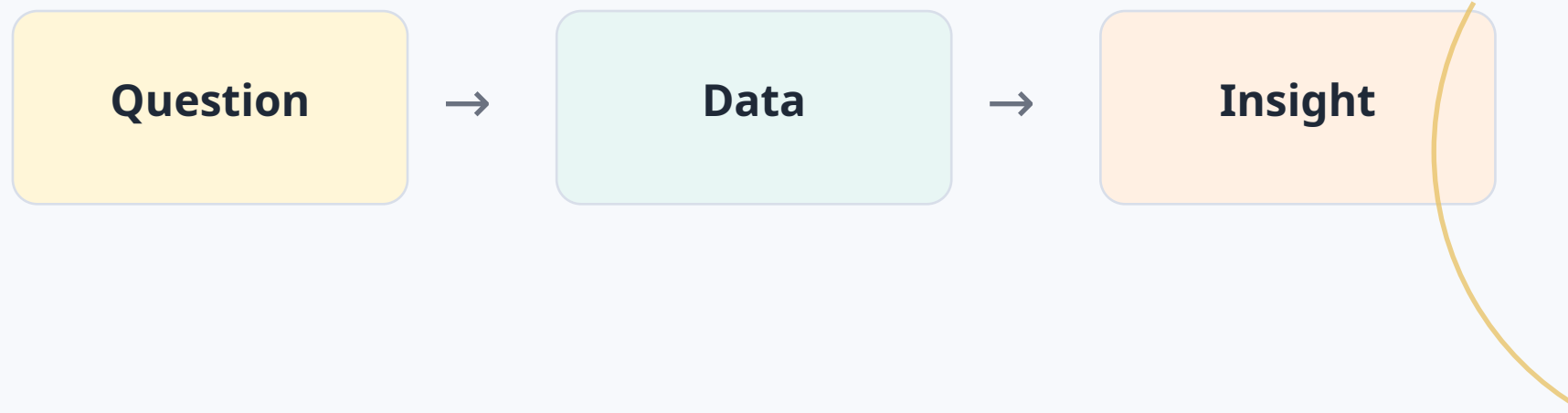


Introduction to Data Analysis with Python and Pandas

From raw data to useful answers



Today's learning path



What is data analysis?



Where datasets come from



Rows, columns and data shapes



The data analysis workflow



Why Python and Pandas?



First Pandas commands

By the end, students should understand the purpose of data analysis, explain the workflow and perform a first dataset inspection.

What is data analysis?

Data analysis is the process of using data to answer questions, solve problems and support decisions.

Data

Raw facts: numbers, words, dates, categories or measurements.

Information

Processed or organised data, such as an average, total or count.

Insight

Meaning that helps us understand what is happening or decide what to do next.

Example: A list of marks becomes information when we calculate the average. It becomes insight when we identify who may need support.

What does a dataset look like?

Rows

Each row is one record or observation: one student, one transaction, one event.

Columns

Each column is one variable or field: mark, programme, date, category.

StudentID	Name	Programme	Mark	Attendance
1001	Aisha	Diploma IT	72	90
1002	Ben	Diploma IT	45	60
1003	Carla	Diploma IS	81	95

Key vocabulary: dataset - row - column - value - variable - observation - data type

Where do datasets come from?

Manual capture

registers, forms, marks

Files

CSV and Excel

Databases

student, sales, hospital systems

Online systems

LMS, websites, apps

Sensors

weather, GPS, devices

Open data

Stats SA, UCI, Kaggle

Before analysing a dataset, ask: Where did it come from? Who created it? When was it collected? What does each column mean?

Before analysis: trust the data?

Source

Where did it come from?

Structure

What does each row and column represent?

Quality

Missing, duplicate or inconsistent values?

Ethics

Does it contain personal information?

Common beginner data issues: inconsistent spelling, blank values, wrong types and duplicate rows.

The data analysis workflow



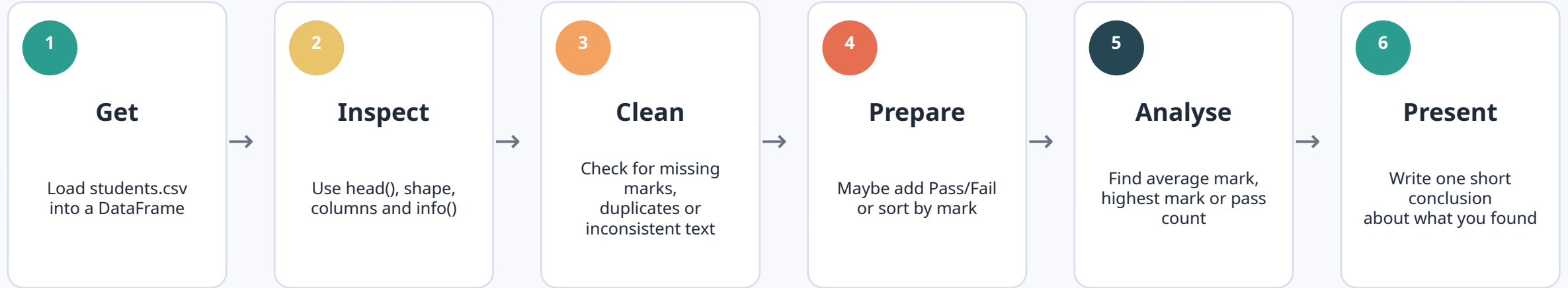
The workflow is not always perfectly linear: analysts often return to earlier steps when they discover a problem.

The workflow in more detail



At each step, keep asking: What problem am I solving, and can I trust the data?

Workflow example: students.csv



This shows that data analysis is a process: we move from raw data to an evidence-based answer.

Workflow checkpoint

Match these actions

Load students.csv
Run df.info()
Remove a duplicate row
Create a PassStatus column
Calculate the average mark
Write a short conclusion

Discussion prompts

Which action belongs to inspection?
Why must inspection come before cleaning?
Could an analyst return to an earlier phase?
Which action communicates the finding?

Use this as a short pair or whole-class activity before the practical exercise.

Why Python and Pandas?

Python

A readable programming language used for automation, analysis, applications and many other tasks.

Pandas

A Python library for table-like data. It helps us load, inspect, clean, analyse and summarise datasets.

DataFrame

The main Pandas structure students will use: a table with rows and columns.

Python is the language. Pandas is the tool we use to work with table-like datasets.

First Pandas commands

```
import pandas as pd

df = pd.read_csv("students.csv")

df.head()      # preview rows
df.shape       # rows and columns
df.info()      # columns, types, missing values
df.describe()  # summary statistics
df["Mark"].mean()
```

import pandas as pd

Load the Pandas library

read_csv

Read a CSV file into a DataFrame

head / shape / info

Inspect before analysing

mean

Answer a simple question

Wide data vs long data

Wide data

Repeated measurements are stored in separate columns. Good for compact reports and human reading.

Student	Test1	Test2	Test3
Aisha	65	70	75
Ben	50	55	60

Long data

Repeated measurements are stored in rows. Often easier for grouping, filtering and charts.

Student	Test	Mark
Aisha	Test1	65
Aisha	Test2	70
Ben	Test1	50

Rule of thumb: wide is often good for reading; long is often good for analysing.

Suggested classroom activities

Activity 1

Identify rows, columns and values in students.csv

Activity 2

Classify dataset sources and possible quality problems

Activity 3

Place common tasks into the correct workflow phase

Activity 4

Rewrite the test marks from wide to long format

Activity 5

Run first Pandas inspection commands and interpret the output

Close with one reflection: Why does inspection come before cleaning?

What students should remember

Data analysis starts with questions, not code.
A dataset is usually a table of rows and columns.
The source and quality of a dataset matter.
The workflow moves from data to insight.
Pandas helps us inspect and analyse table-like data.
Next lesson: selecting, filtering and sorting data.

One final question for students: What question did we answer using the data today?